Comprehensive Literature Searching

Adhering to Reporting Guidelines
&
Maintaining Awareness of Flawed Science



Amy Riegelman, aspringe@umn.edu
Social Sciences Librarian



Wilson Library

Collections

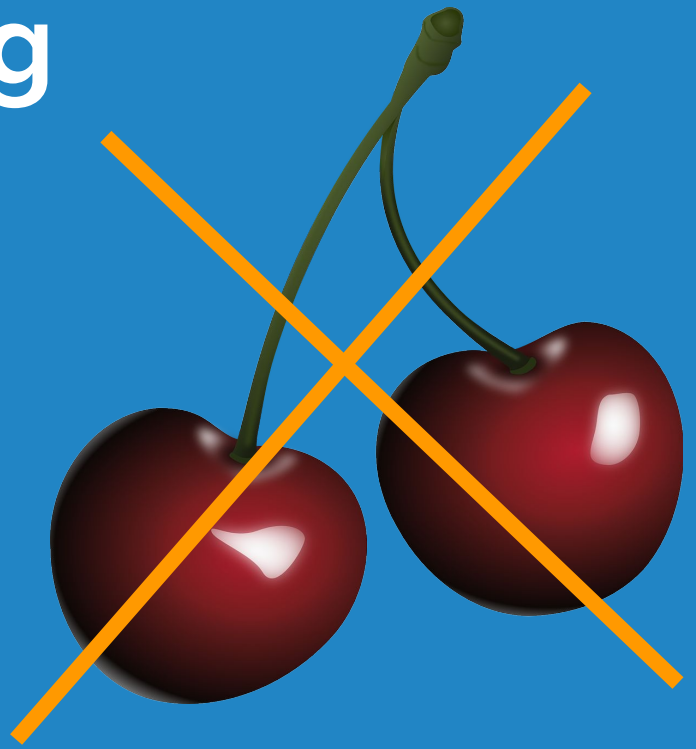
- arts humanities
- social sciences
- special collections

Spaces

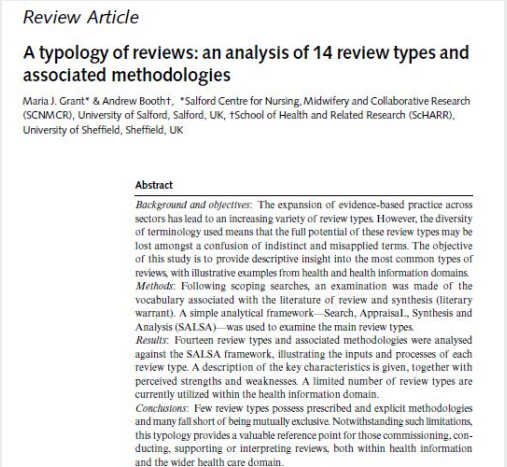
- Collaboration Studio
- SMART Learning Commons
- Meeting rooms
- Labs
- Dunn Bros.

Comprehensive Literature Searching

What does this mean?



Literature Reviews



Review Article

A typology of reviews: an analysis of 14 review types and associated methodologies

Maria J. Grant* & Andrew Booth†, *Salford Centre for Nursing, Midwifery and Collaborative Research (SCNMCR), University of Salford, Salford, UK, †School of Health and Related Research (ScHARR), University of Sheffield, Sheffield, UK

Abstract

Background and objectives: The expansion of evidence-based practice across sectors has lead to an increasing variety of review types. However, the diversity of terminology used means that the full potential of these review types may be lost amongst a confusion of indistinct and misapplied terms. The objective of this study is to provide descriptive insight into the most common types of reviews, with illustrative examples from health and health information domains.

Methods: Following scoping searches, an examination was made of the vocabulary associated with the literature of review and synthesis (literary warrant). A simple analytical framework—Search, AppraisaL, Synthesis and Analysis (SALSA)—was used to examine the main review types.

Results: Fourteen review types and associated methodologies were analysed against the SALSA framework, illustrating the inputs and processes of each review type. A description of the key characteristics is given, together with perceived strengths and weaknesses. A limited number of review types are currently utilized within the health information domain.

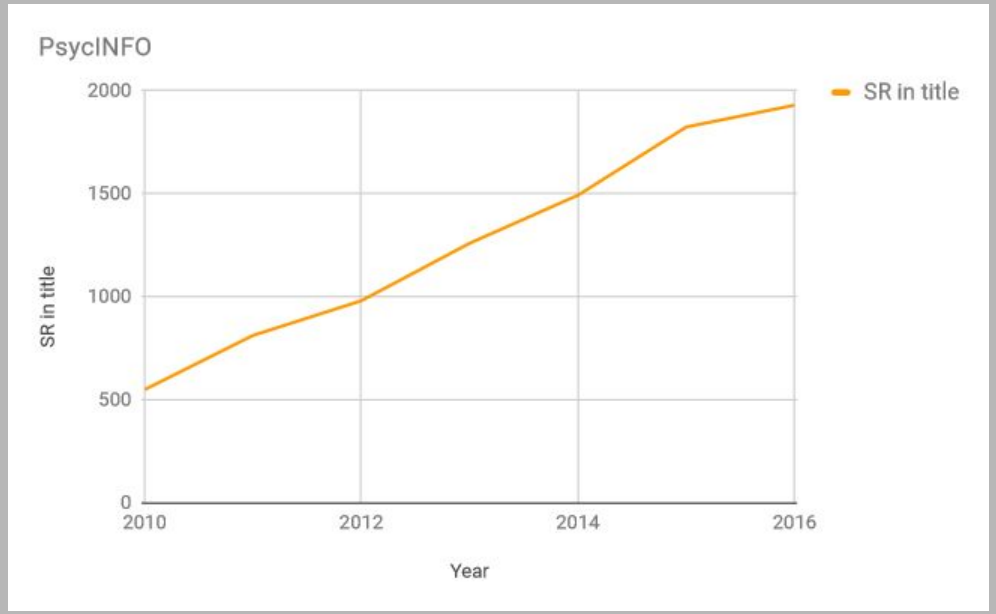
Conclusions: Few review types possess prescribed and explicit methodologies and many fall short of being mutually exclusive. Notwithstanding such limitations, this typology provides a valuable reference point for those commissioning, conducting, supporting or interpreting reviews, both within health information and the wider health care domain.

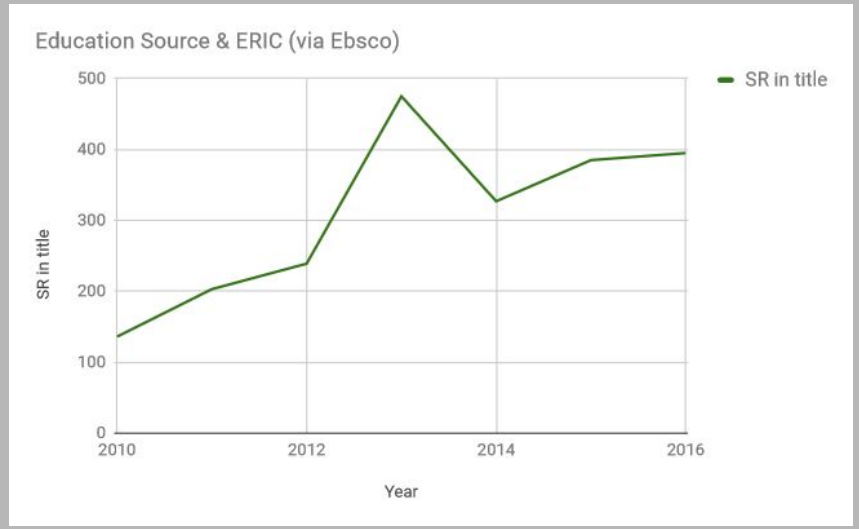
"published materials that provide examination of recent or current literature. Can cover wide range of subjects at various levels of completeness and comprehensiveness" (Grant & Booth, 2009)

Goals



The numbers: based on a title search for “systematic review” OR “meta-analysis”





Systematic Review

What is a systematic review?

A systematic review is a research method in which a team formulates a research question, searches, selects, and appraises the literature in order to test and evaluate in order for researchers, practitioners, and policy-makers to make evidence based decisions.

Unlike other types of reviews, this research method includes a reproducible and transparent methodology. For help differentiating between the various types of reviews (e.g., literature review, scoping review), consult *A Typology of Reviews* (Grant & Booth, 2009).

Systematic Review Service Request

How can the libraries help?

Librarians have skills to assist you throughout the process especially in helping you develop and report a comprehensive and reproducible search strategy.

1: Planning

- Guidance on SR process and steps
- Background searching for current and upcoming reviews
- Help with development or refinement of review topic

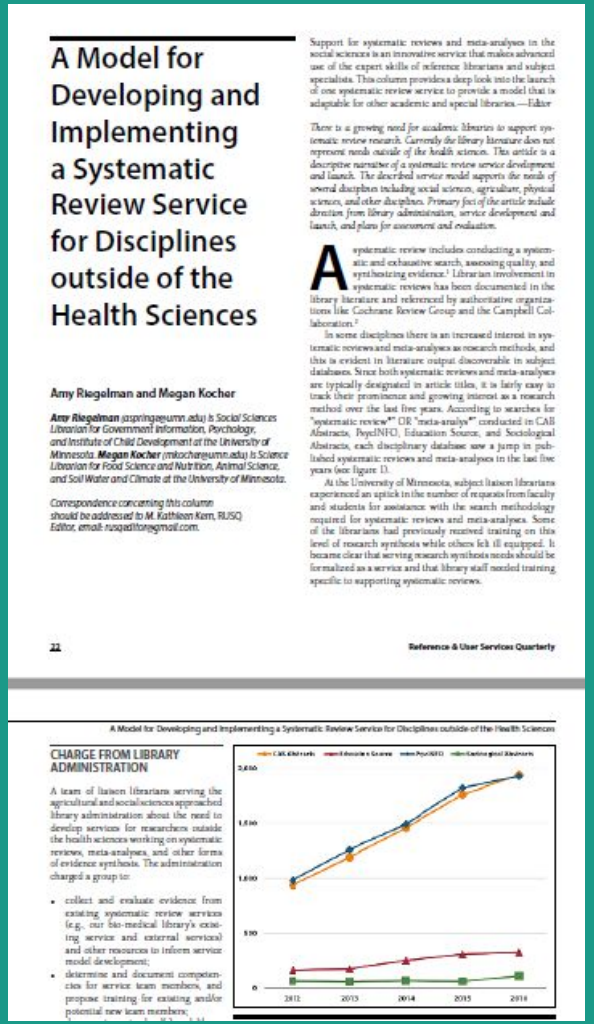
2: Searching

- Identification of databases for searches
- Development and execution of searches
- Documentation of search strategies
- Management of search results
- Guidance on methods for study review and data management

3: Reporting

- Drafting of literature search description in methods section
- Drafting of literature search appendix
- Guidance for complying with reporting guidelines (e.g., PRISMA)
- Review of manuscript drafts

Riegelman, A. & Kocher, A. (2018). A Model for Developing and Implementing a Systematic Review Service for Disciplines Outside of the Health Sciences. *Reference User Services Quarterly*, 58(1), 22-27. doi: 10.5860/rusq.58.1.6837



A Model for Developing and Implementing a Systematic Review Service for Disciplines outside of the Health Sciences

Amy Riegelman and Megan Kocher

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Support for systematic reviews and meta-analyses in the social sciences is an innovative service that makes advanced use of the expert skills of reference librarians and subject specialists. This column provides a deep look into the launch of one systematic review service to provide a model that is adaptable for other academic and special libraries.—Editor

There is a growing need for academic libraries to support systematic review research. Currently the library literature does not represent needs outside of the health sciences. This article is a descriptive narrative of a systematic review service development and launch. The described service model supports the needs of several disciplines including social sciences, agriculture, physical sciences, and other disciplines. Primary foci of the article include direction from library administration, service development and launch, and plans for assessment and evaluation.

A systematic review includes conducting a systematic and exhaustive search, assessing quality, and synthesizing evidence.[1] Librarian involvement in systematic reviews has been documented in the library literature and referenced by authoritative organizations like Cochrane Review Group and the Campbell Collaboration.[2]

In some disciplines there is an increased interest in systematic reviews and meta-analyses as research methods, and this is evident in literature output discoverable in subject databases. Since both systematic reviews and meta-analyses are typically designated in article titles, it is fairly easy to track their prominence and growing interest as a research method over the last five years. According to searches for "systematic review*" OR "meta-analys*" conducted in CAB Abstracts, PsycINFO, Education Source, and Sociological Abstracts, each disciplinary database saw a jump in published systematic reviews and meta-analyses in the last five years (see figure 1).

At the University of Minnesota, subject liaison librarians experienced an uptick in the number of requests from faculty and students for assistance with the search methodology required for systematic reviews and meta-analyses. Some of the librarians had previously received training on this level of research synthesis while others felt ill equipped. It became clear that serving research synthesis needs should be formalized as a service and that library staff needed training specific to supporting systematic reviews.

CHARGE FROM LIBRARY ADMINISTRATION

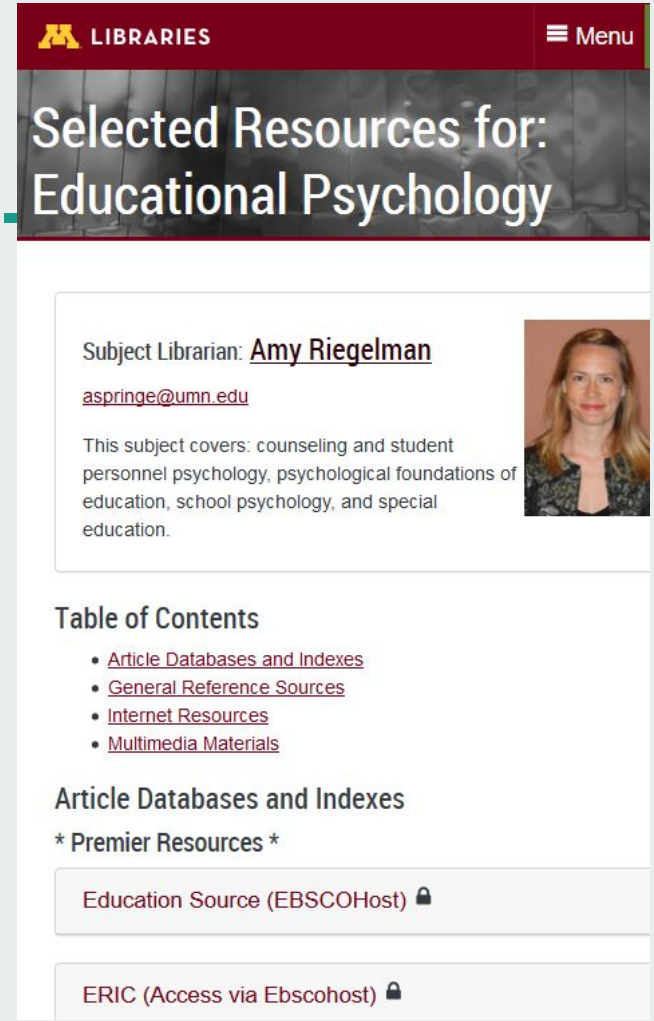
A team of liaison librarians serving the agricultural and social sciences approached library administration about the need to develop services for researchers outside the health sciences working on systematic reviews, meta-analyses, and other forms of evidence synthesis. The administration charged a group to:

- collect and evaluate evidence from existing systematic review services (e.g., our bio-medical library's existing service and external services) and other resources to inform service model development;
- determine and document competencies for service team members, and propose training for existing and/or potential new team members;

Examples of research synthesis*	Reference to expert searching, involvement of information specialist, or similar concept
Campbell (systematic) reviews	The Campbell Collaboration states that "[r]elevant methodological expertise includes: **information/library science (searching and text retrieval)**"
Cochrane (systematic) reviews	"If a [Cochrane Review Group] is currently without a Trials Search Co-ordinator authors should seek the guidance of a local **healthcare librarian or information specialist**, where possible one with experience of conducting searches for systematic reviews." (Lefebvre, Manheimer, and Glanville 6.1.1.1)
Effectiveness and comparative effectiveness reviews	In "Finding Evidence for Comparing Medical Interventions" published by the Agency for Healthcare Research and Quality (AHRQ, US), Relevo and Balshem state that "[a] **librarian or other expert searcher** should be involved in the development of the search"

Where to search





z.umn.edu/epsy

Where to search for peer-reviewed articles

Subject databases

- Education Source
- ERIC
- PsycINFO
- SPORT Discus

Multidisciplinary databases

- Web of Science
- Scopus
- Academic Search Premier
- Google Scholar

General tips for searching

- **Phrasing** e.g., “educational psychology,” “special education,”
- **Wildcard** e.g., psych* or counsel*
- **Boolean operators** AND, OR, NOT e.g., ASD OR Autis* AND "therap*" NOT “Acute Stress Disorder”

Search records & fields

What are you actually searching?

- Title
- Publication date
- Language
- Author
- Journal/Source
- ISSN
- Publication type
- Abstract
- DOI
- Author provided keywords
- Subject headings

Controlled Vocabulary

Subject Headings

Subject Descriptors

Include Synon

Back to Search Results

Condoms Distribution (2004) Synonym

Instead of this term use:

Sexually Transmitted Diseases

ED.gov ies INSTITUTE OF EDUCATION SCIENCES

Scope Note for: *Test Anxiety*

MAIN TERM: Test Anxiety

DATE OF ENTRY: 1967

SCOPE NOTE:

Fear or tension in anticipation of

How do we find grey lit?

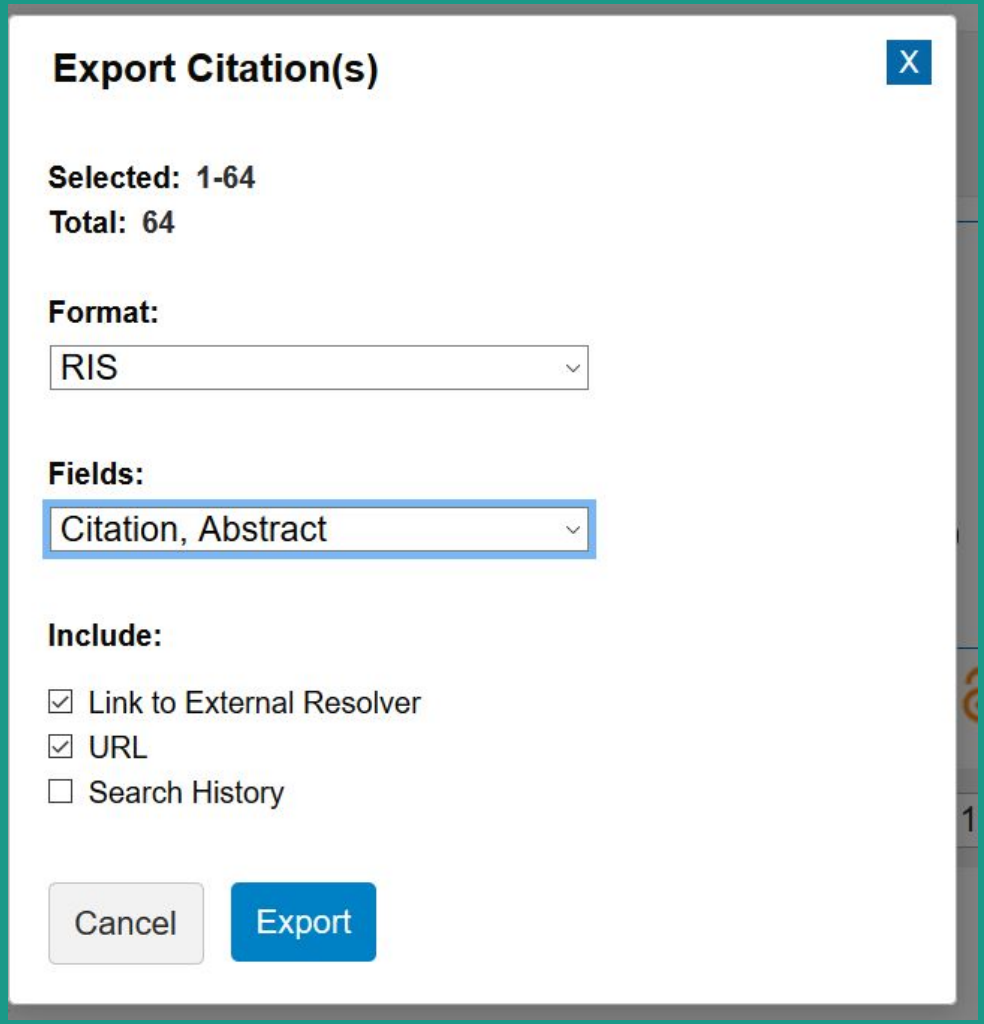
- Library databases (sometimes!)
- Preprint repositories
  - OSF Preprints
  - Preprints.org
  - SSRN
- Proquest Digital Theses/Dissertations

Citation Managers







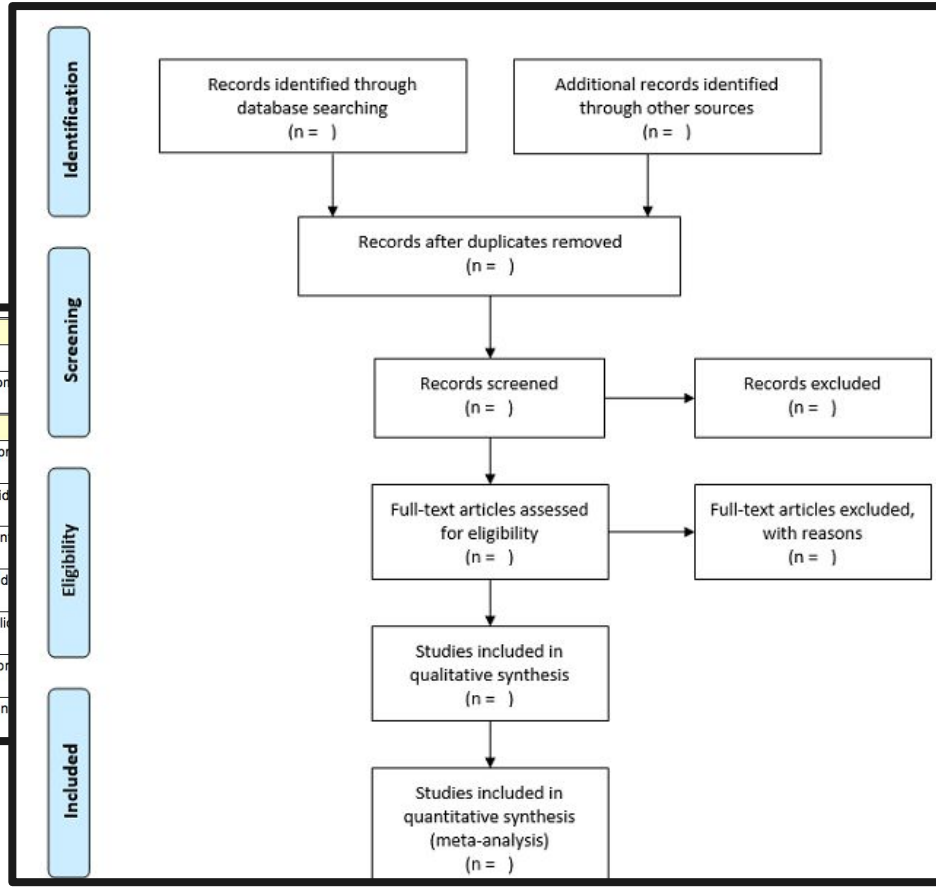


Exporting results to a citation manager

PRISMA

Reporting Guidelines

INTRODUCTION		
Rationale	3	Describe the rationale for the review in the context of what is already known.
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, cor outcomes, and study design (PICOS).
METHODS		
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, pr registration information including registration number.
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years consid language, publication status) used as criteria for eligibility, giving rationale.
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to iden additional studies) in the search and date last searched.
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could repeated.
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if appli included in the meta-analysis).
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any pr for obtaining and confirming data from investigators.
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumption simplifications made.



Before you submit your manuscript..

.. check to make sure that the research hasn't been retracted or corrected.

6.4.10 Identifying fraudulent studies, other retracted publications, errata and comments

When considering the eligibility of studies for inclusion in a Cochrane review, it is important to be aware that some studies may have been found to be fraudulent or may for other reasons have been retracted since publication. Reports of studies indexed in MEDLINE that have been retracted (as fraudulent or for other reasons) will have the Publication Type term 'Retracted Publication' added to the record. The article giving notice of the retraction will have the Publication Type term 'Retraction of Publication' assigned. Prior to any decision being taken to retract an article, articles may be published that refer to an original article and raise concerns of this sort. Such articles would be classified as a Comment. The US National Library of Medicine's (NLM's) policy on this is that "Among the types of articles that will be considered comments are: ….. announcements or notices that report questionable science or investigations of scientific misconduct (sometimes published as 'Expression of concern')".

Retraction: The action of withdrawing a statement, accusation, etc., which is now admitted to be erroneous or unjustified; disavowal; recantation; an instance of this; a statement making such a withdrawal.

retraction, n. (2016, March). *OED Online*. Oxford University Press

How do we find out about retractions?

- Journals publish a notice
- Retraction Watch (blog)
- News sources (e.g., Wakefield's now retracted autism-vaccine article)
- Ideally: notice is featured in library databases or aggregators



The following article has been retracted by the Editor

The New York Times

WORLD | U.S. | N.Y. / REGION | BUSINESS | TECHNOLOGY | SCIENCE | HEALTH | SPORTS | OPINION

Study Linking Vaccine to Autism Is Called Fraud

Published: January 6, 2011

LONDON (AP) -- The first study to link a childhood vaccine to autism was based on doctored information about the children involved, according to a new report on the widely discredited research.

The conclusions of the 1998 paper by Dr. Andrew Wakefield and his colleagues were renounced by 10 of its 13 authors and later retracted by the medical journal The Lancet, where it was published. Still, the suggestion that the combined measles, mumps and rubella vaccine was connected to autism rattled parents worldwide; immunization rates for the MMR shot have never fully recovered.

FACEBOOK
TWITTER
GOOGLE+
EMAIL
SHARE
PRINT
REPRINTS

A new examination found, by comparing the reported diagnoses in the paper with hospital records, that Dr. Wakefield and his colleagues had altered facts about patients in their study.

HONEST ERROR

Mahajan, N., Barnes, J. L., Blanco, M., & Santos, L. R. (2009). Enumeration of objects and substances in non-human primates: Experiments with brown lemurs (Eulemur fulvus). *Developmental Science*, *12*(6), 920-928.

CITED 13x

MISCONDUCT

Stapel, D., Suls, J., & Dovidio, John F. (2004). Method matters: Effects of explicit versus implicit social comparisons on activation, behavior, and self-views. *Journal of Personality and Social Psychology*, *87*(6), 860-875.

CITED 54x

Reasons for Retraction

- 21.3% error
- 67.4% misconduct
  - 43.4% fraud
  - 14.2% duplicate publication
  - 9.8% plagiarism

Fang, Steen & Casadevall, 2012

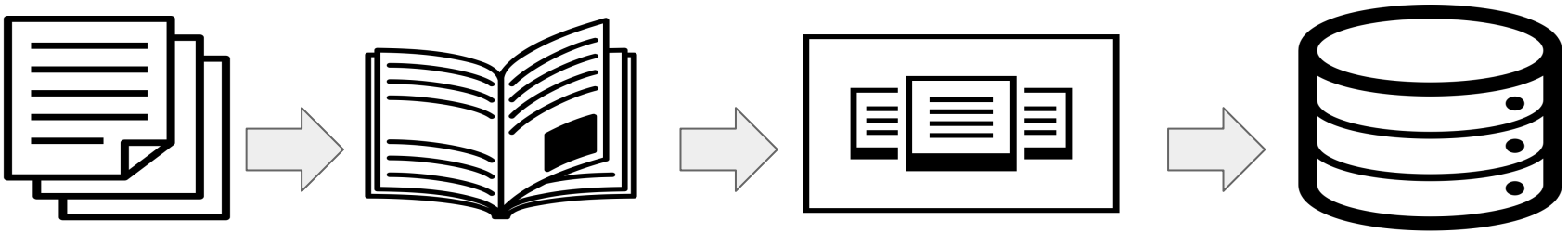
2% to 4% of researchers admit having fabricated, falsified, or altered data or results

Fanelli, D. (2009) How many scientists fabricate and falsify research? A systematic review and meta-analysis of survey data. PLoS One, 4(5), e5738.

Discovering Retractions while using Library tools

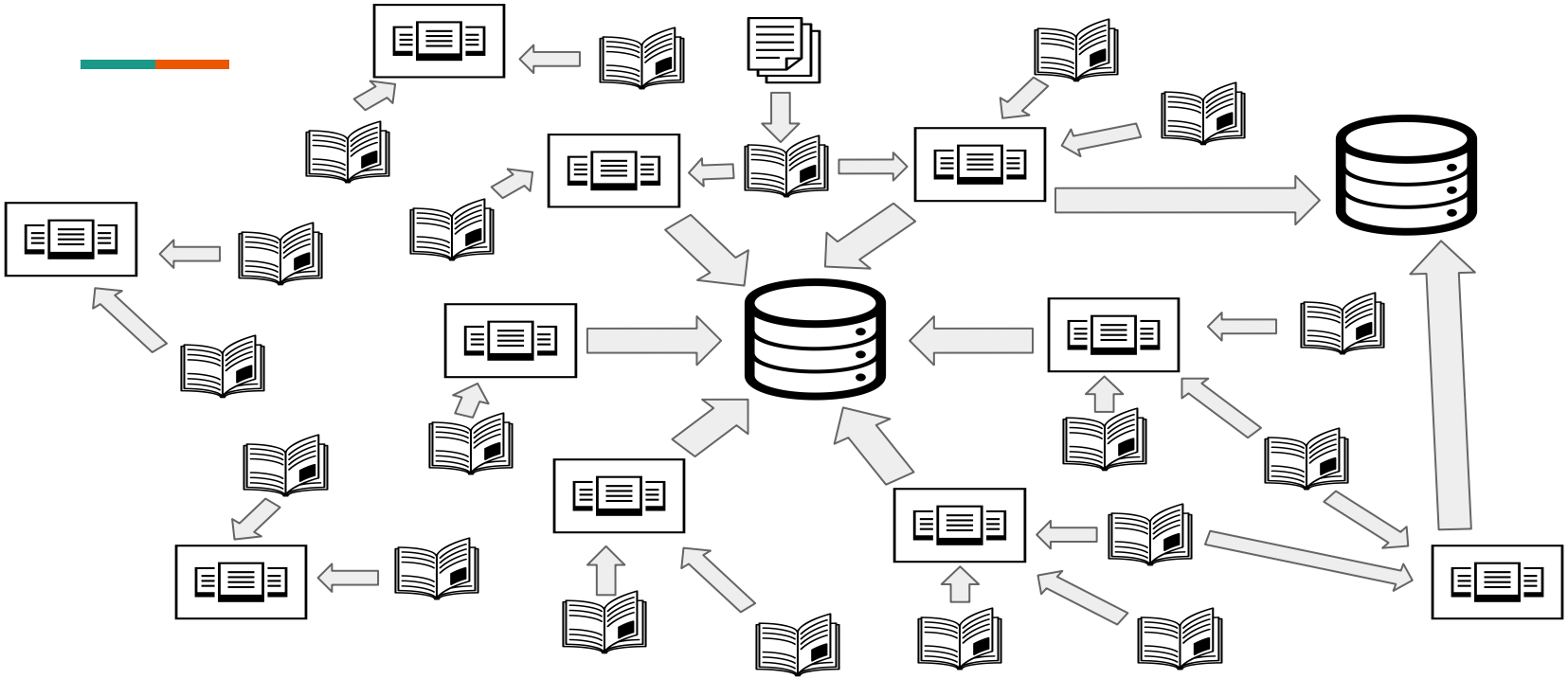


The Life of an Article



Article Journal Database Aggregator

Images: Document by Michael Loupos, Book by Chris Homan, Carousel by Lil Squid, Database by Shmidt Sergey



European Child and Adolescent Psychiatry
Volume 12, Issue 3, 2003, Pages 144-148

Acamprosate and its efficacy in treating alcohol dependent adolescents

Niederhofer, H. , Staffen, W.
Christian Doppler-Klinik, 5020 Salzburg, Austria

Abstract

Background: About 50% of adult alcoholic patients relapse within 3 months of treatment. Previous studies to assess the **efficacy** and safety of long-term **acamprosate** treatment in **alcohol** dependence of adol 16-19 years, with chronic or episodic **alcohol** dependence. Patients were randomly allocated treatment treatment started and on days 30, and 90 by interview, self report, questionnaire, and laboratory screeni phase: of those withdrawn, 11 (1 vs 6) relapsed, 5 (3 vs 2) refused to continue treatment, 3 (1 vs 2 **acamprosate** treated and 2 placebo-treated patients had been continuously abstinent (p = 0.0076). Mea placebo group (79.8 [SD 37.5] vs 32.8 [19.0] days; p= 0.012). Interpretation: **Acamprosate** is an effectiv

Author keywords

Acramposate; **Adolescents**; **Alcohol**

Indexed keywords

EMTREE drug terms: acamprosate

EMTREE medical terms: abstinence; adolescent; adult; alcoholism; article; clinical article; clinical trial; laboratory test; male; questionnaire; randomized controlled trial; side effect; treatment outcome

MeSH: Adolescent; Adolescent Behavior; Adult; **Alcohol** Deterrents; Alcoholism; Double-Blind Method; F
Medline is the source for the MeSH terms of this document.

Chemicals and CAS Registry Numbers: acamprosate, 77337-73-6;acamprosate, 77337-76-9; Alcoho

ISSN: 10188827 CODEN: EAPSE Source Type: Journal Original language: English
DOI: 10.1007/s00787-003-0327-1 PubMed ID: 12768462 Document Type: Article

RETRACTED ARTICLE
See: Retraction Notice

Eur Child Adolesc Psychiatry. 2003 Jun;12(3):144-8.

Acamprosate and its efficacy in treating alcohol dependent adolescents.

Niederhofer H[1], Staffen W.
Author information

Retraction in
Retraction note: Acamprosate and its efficacy in treating alcohol dependent adolescents. [Eur Child Adolesc Psyc

Abstract

BACKGROUND: About 50 % of adult alcoholic patients relapse within 3 months of treatment. Previous studies h acamprosate may help to prevent such relapse. The aim of our study was to assess the efficacy and safety of lo treatment in alcohol dependence of adolescents.

METHODS: In this, double-blind, placebo-controlled study, we recruited 26 patients, aged 16-19 years, with chro dependence. Patients were randomly allocated treatment with acamprosate (1332 mg daily) or placebo for 90 d on the day treatment started and on days 30, and 90 by interview, self report, questionnaire, and laboratory scre

FINDINGS: 13 acamprosate-treated and 13 placebo-treated patients completed the treatment phase: of those w 5 (3 vs 2) refused to continue treatment, 3 (1 vs 2) had concurrent illness, and 2 (1 vs 1) had adverse side-effec acamprosate treated and 2 placebo-treated patients had been continuously abstinent (p = 0.0076). Mean cumul significantly greater in the acamprosate group than in the placebo group (79.8 [SD 37.5] vs 32.8 [19.0] days; p =

INTERPRETATION: Acamprosate is an effective and well-tolerated pharmacological adjunct to psychosocial trea

PMID: 12768462 DOI: 10.1007/s00787-003-0327-1
[PubMed - indexed for MEDLINE]

Publication Types, MeSH Terms, Substances

LinkOut - more resources

What We Did

- Examine a study of retracted mental health literature identified through Retraction Watch
- Document how these retractions are represented across a variety of resources relevant to the discipline

Publisher sites

Medline via OVID

PsycINFO via OVID

EBSCOhost

Scopus

Web of Science

Medine via PubMed

Out of these 812 records, 487 (60%) indicated that the article had been retracted

Bakker, C., & Riegelman, A. (2018). Retracted Publications in Mental Health Literature: Discovery across Bibliographic Platforms. *Journal of Librarianship and Scholarly Communication*, 6(1). DOI: 10.7710/2162-3309.2199



RESEARCH

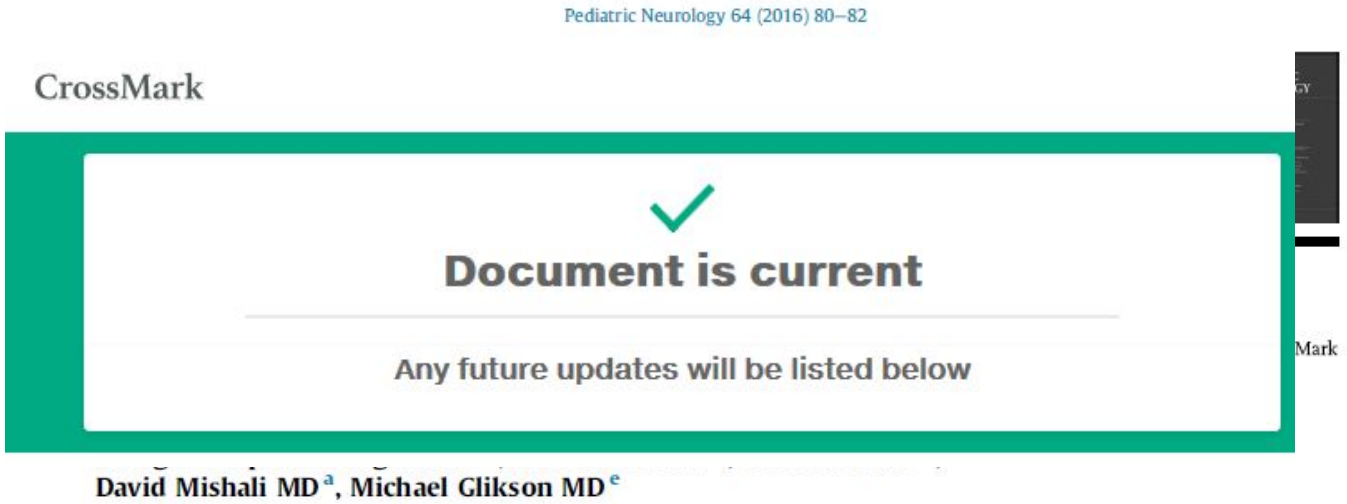
Retracted Publications in Mental Health Literature: Discovery across Bibliographic Platforms

Caitlin Bakker
Biomedical/Research Services Liaison, University of Minnesota
Amy Riegelman
Social Sciences Librarian, University of Minnesota

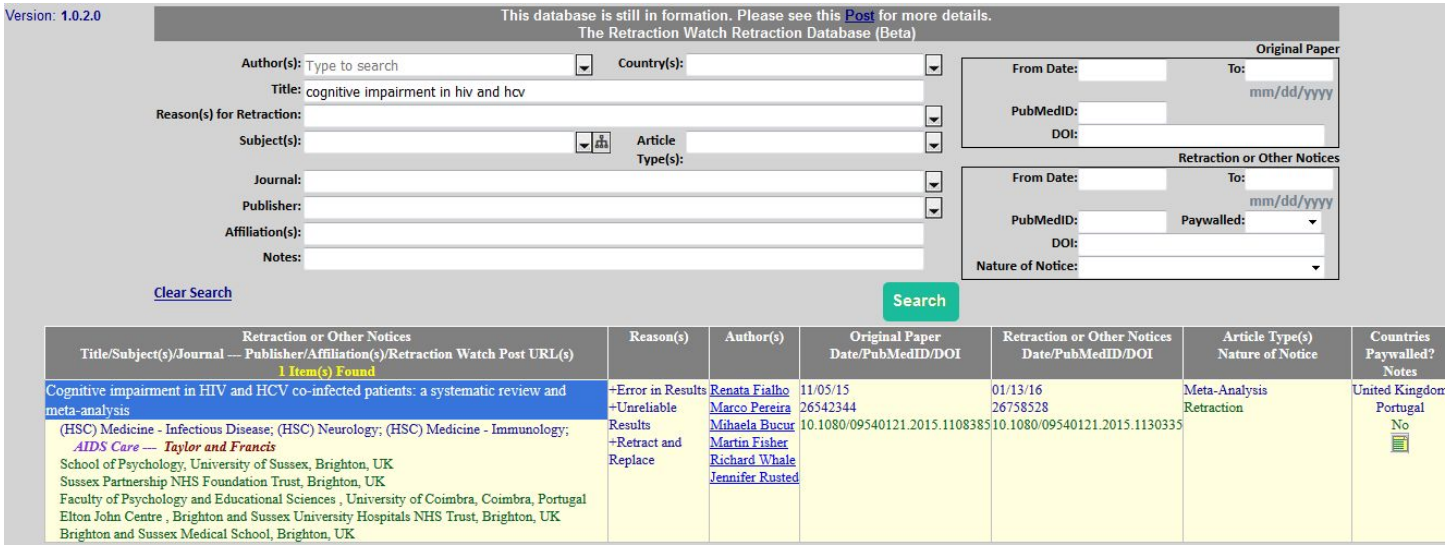
INTRODUCTION Retractions are a mechanism by which science corrects itself, withdrawing statements or claims that have proven to be erroneous. However, this requires that such corrections be displayed clearly and consistently. This paper considers how retracted publications in the mental health literature are represented across different platforms. **METHODS** Using Retraction Watch, we identified 144 retracted articles in the mental health field. We looked across seven platforms to determine the consistency and clarity of the retracted status of these publications. **RESULTS** Of the 812 records for retracted publications, 40.0% (n=325) did not indicate that the paper had been retracted. Of available PDFs, 26.3% (53/201) did not indicate that the paper had been retracted. Of the 144 articles studied, only 10 were represented as being retracted across all resources through which they were available. **DISCUSSION** Retracted publications in this sample were inconsistently represented across library resources. While technical solutions, such as Crossmark by Crossref, may help mitigate these challenges, the inconsistent display of retractions has implications for education and outreach. **CONCLUSION** Our study found that the retractions in our sample were not clearly and consistently represented across sources. Libraries, which provide access to and training in these resources, have a responsibility to raise awareness of these inconsistencies and to advocate for more timely and accurate metadata.

Tools for discovering corrections and retractions

Crossmark


Pediatric Neurology 64 (2016) 80–82
CrossMark
Document is current
Any future updates will be listed below
David Mishali MD [a], Michael Glikson MD [e]

RetractionDatabase.org



Version: 1.0.2.0

This database is still in formation. Please see this Post for more details.
The Retraction Watch Retraction Database (Beta)

Author(s): Type to search
Country(s):
Title: cognitive impairment in hiv and hcv
Reason(s) for Retraction:
Subject(s):
Article Type(s):
Journal:
Publisher:
Affiliation(s):
Notes:

Original Paper
From Date: To: mm/dd/yyyy
PubMedID:
DOI:

Retraction or Other Notices
From Date: To: mm/dd/yyyy
PubMedID: Paywalled:
DOI:
Nature of Notice:

Clear Search
Search

Retraction or Other Notices Title/Subject(s)/Journal --- Publisher/Affiliation(s)/Retraction Watch Post URL(s)	Reason(s)	Author(s)	Original Paper Date/PubMedID/DOI	Retraction or Other Notices Date/PubMedID/DOI	Article Type(s) Nature of Notice	Countries Paywalled? Notes
1 Item(s) Found						
Cognitive impairment in HIV and HCV co-infected patients: a systematic review and meta-analysis (HSC) Medicine - Infectious Disease; (HSC) Neurology; (HSC) Medicine - Immunology; *AIDS Care* --- *Taylor and Francis* School of Psychology, University of Sussex, Brighton, UK Sussex Partnership NHS Foundation Trust, Brighton, UK Faculty of Psychology and Educational Sciences , University of Coimbra, Coimbra, Portugal Elton John Centre , Brighton and Sussex University Hospitals NHS Trust, Brighton, UK Brighton and Sussex Medical School, Brighton, UK	+Error in Results +Unreliable Results +Retract and Replace	Renata Fialho Marco Pereira Mihaela Bucur Martin Fisher Richard Whale Jennifer Rusted	11/05/15 26542344 10.1080/09540121.2015.1108385	01/13/16 26758528 10.1080/09540121.2015.1130335	Meta-Analysis Retraction	United Kingdom Portugal No

Questions?

Amy Riegelman

aspringe@umn.edu

LIBRARIES
UNIVERSITY OF MINNESOTA